

Henrique Baptista

ML / AI Engineer | Production AI Systems & Evaluation

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PROFESSIONAL SUMMARY

ML / AI Engineer building reliable production AI systems across healthcare and SaaS. Hands-on ownership from data curation and model evaluation to Python APIs, deployment, and user-facing products. Specialized in LLM quality, Voice AI/ASR, and measurable reliability through benchmarking, failure analysis, monitoring, and cost/latency optimization.

PROFESSIONAL EXPERIENCE

ML / AI Engineer - VOA Health

Feb 2025 – Present

São Paulo, Brazil · HealthTech, AI for medical documentation and clinical workflows

LLM Quality & Evaluation

- Built an automated QA pipeline combining heuristic detectors with LLM-as-Judge to evaluate AI-generated clinical documents and catch known failure patterns before release
- Created a root-cause bug catalog and production sampling methodology to estimate real-world error rates, guide targeted improvements, and surface cost optimization opportunities

Data Curation & ML Pipelines

- Designed an end-to-end data curation pipeline with dual-annotator consensus, processing thousands of domain-specific samples for model training and evaluation
- Built a modular 8-component Hugging Face ↔ Argilla ↔ AWS S3 workflow with Docker deployment, monitoring, duplicate detection, and reviewer adjudication

Voice AI & ASR Benchmarking

- Benchmarked 9 commercial ASR providers using WER/CER, latency, and medical terminology analysis; cataloged 217 error variations across the 10 most error-prone medications
- Optimized VAD parameters and Portuguese normalization for noisy clinical audio, reducing false segment cuts and improving downstream transcription quality

PROJECTS

WordinAI - SaaS Transcription & AI Document Generation

Next.js · LangGraph · Whisper · Supabase

- Built a production SaaS with LangGraph agent workflows, tool calling, semantic transcript search, and multi-provider STT for automated document generation
- Designed multi-provider routing with circuit breakers and retries to optimize transcription latency, cost, and resilience across external services

Datathon Passos Mágicos - Education Risk Prediction

XGBoost · FastAPI · Evidently AI

- Built an end-to-end ML system to identify educational delay risk for an NGO, achieving **92.59% accuracy / 0.97 ROC-AUC** with API, dashboard, drift monitoring, Docker, and Hugging Face Spaces deployment

EDUCATION

Postgraduate Program, Machine Learning Engineering - FIAP

Mar 2025 – Mar 2026

Bachelor, Systems Analysis & Development - FIAP

Feb 2023 – Dec 2024

TECHNICAL SKILLS

LLM & AI Quality: LangGraph, LangChain, OpenAI API, LLM-as-Judge, prompt engineering, agent workflows, evaluation datasets

ML & Data: PyTorch, Hugging Face (Transformers, Datasets, Hub), Whisper, WER/CER benchmarking, data curation, Argilla, XGBoost, computer vision

Engineering: Python, FastAPI, TypeScript, Next.js, PostgreSQL, Supabase, Docker, CI/CD, Git, AWS S3

Production AI: Voice AI (STT/ASR, VAD), model evaluation, monitoring, multi-provider routing, document generation, failure analysis

Languages: Portuguese (Native) · English (Fluent)